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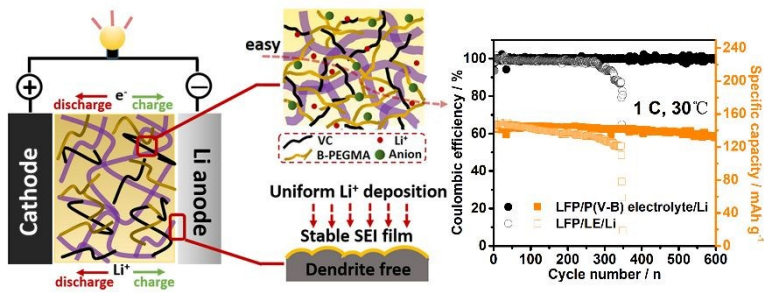
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TOC

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A borate decorated anion-immobilized solid polymer electrolyte enables stable solid electrolyte interface film with uniform  $\text{Li}^+$  deposition and long-term cycling stability for dendrite-free lithium metal batteries.



# Borate decorated anion-immobilized solid polymer electrolyte for dendrite-free, long-life Li metal batteries

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## Abstract

Abrupt Li dendrite growth and the safety hazards caused by liquid electrolytes are generally acknowledged as major technical barriers for practical application of Li metal batteries. Solid polymer electrolytes (SPEs), are promising to overcome these obstacles, but suffer from the rigidity–conductivity inconsistency, ununiform ion distribution and inferior interfacial compatibility. Herein, an anion-immobilized SPE using vinylene carbonate as the rigid polymer backbone and flexible ether oxygen chains containing anion-trapping boron moieties is proposed, which facilitates to the Li<sup>+</sup> transport and adjusts the ion distribution. This ingenious design along with facile in-situ preparation effectively integrates a favorable Young's modulus (2.41 GPa), high ionic conductivity ( $9.11 \times 10^{-4}$  S cm<sup>-1</sup> at 25 °C) and a high Li<sup>+</sup> transference number (0.68), as well as achieves a stable solid electrolyte interface layer. As a result, these

integrative properties enable dendrite-free  $\text{LiFePO}_4/\text{Li}$  batteries with excellent rate capacity (8 C, 98.3 mAh g<sup>-1</sup>) and superior long-term cyclability over 600 cycles at 30 °C, providing a new strategy for safe and high-energy all-solid-state energy storage systems.

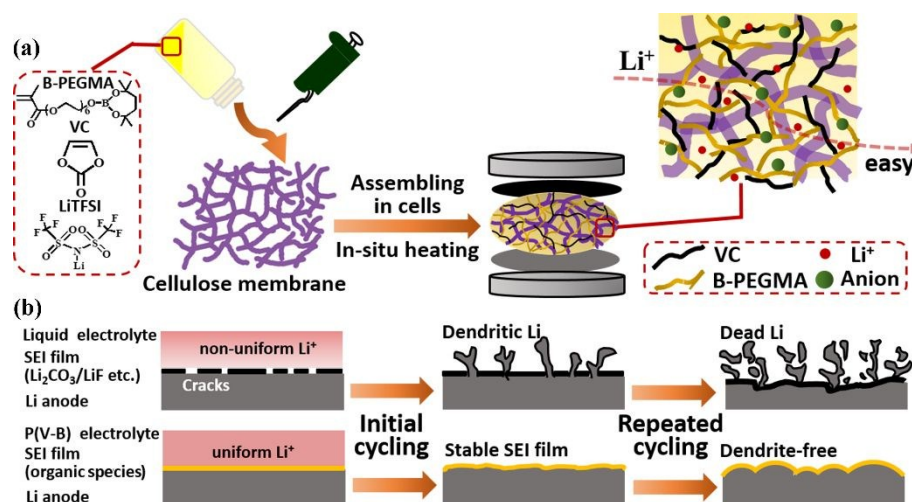
## Introduction

Rechargeable batteries with the virtues of high safety, high-energy density and low-cost are urgently needed to meet the expanding demands for future energy storage systems<sup>1, 2</sup>. Li metal batteries (LMBs) have attracted much attention as the next generation advanced battery technology due to the high theoretical specific capacity (3860 mAh g<sup>-1</sup>), low density (0.534 g cm<sup>-3</sup>) and the lowest reduction potential (-3.040 V vs. the standard hydrogen electrode (SHE)) of Li metal<sup>3-5</sup>. However, potential issues such as short lifetime and short-circuiting, which result from the continuous growth of Li dendrites during repetitive Li plating and stripping, impede significant development of LMBs<sup>5-7</sup>. Some advances in preventing Li dendrite growth have been achieved via optimizing the chemical components of electrolytes<sup>8, 9</sup>, engineering functional separators<sup>10, 11</sup>, artificial fabrication of solid electrolyte interface (SEI) layers<sup>12, 13</sup>, and constructing three-dimensional (3D) current collectors<sup>14, 15</sup>, etc. However, most efforts are limited by intrinsic safety concerns, ranging from uncontrolled side reactions at the Li anode interface to underlying risks (leakage, fires and explosions) related to liquid electrolytes (LEs)<sup>16</sup>.

Solid state electrolytes, especially the solid polymer electrolytes (SPEs), present a promising route to enhancing the safety performance and may allow for dendrite-free all-solid-state LMBs owing to their high thermal and electrochemical stability, as well as the reduced side reactions with Li metal anode<sup>17, 18</sup>. Despite the progresses achieved in mechanically blocking dendrite growth by SPEs with moderate mechanical strength<sup>19, 20</sup>, it's difficult to avoid the penetration of dendritic Li merely by improving the shear modulus, owing to the inevitable uneven electrodeposition behavior of Li. Chazalviel et al. revealed that the anion depletion near the Li electrode leads to a space-charge region that accounts for ramified depositions<sup>5, 21</sup>, while

anion-immobilized electrolyte with high ionic conductivity can facilitate regulating the ion distribution to achieve uniform Li deposits<sup>22, 23</sup>. Besides, the interfacial chemistry stability relevant to the SEI also has a great effect on initial dendrite seeds and inhomogeneous nucleation<sup>3, 24, 25</sup>. Nevertheless, it still remains a great challenge to effectively design SPEs with all these excellent characteristics of reliable mechanical strength, high ion mobility to regulate Li depositing behavior, and be feasible to form a stable interface with the lithium anode.

From this perspective, we have designed an anion-immobilized solid state polymer electrolyte (denoted as P(V-B)) via a facile in-situ, one step polymerization of rigid vinylene carbonate (VC) and flexible poly(ethylene glycol) methyl ether methacrylate containing cyclic boroxane groups (B-PEGMA) on a cellulose membrane framework, as illustrated in Scheme 1a. The as-prepared flexible P(V-B) electrolyte presents desirable physical mechanical and electrochemical properties, including: (i) good mechanical stability with an average Young's modulus of 2.41 GPa; (ii) high ionic conductivity of  $9.11 \times 10^{-4} \text{ S cm}^{-1}$  at 25 °C, resulting from the enhanced interaction with Li salts and segmental mobility by the ether oxygen (EO) moieties; and (iii) high  $\text{Li}^+$  transference number of 0.68 and exceptional interfacial stability against random  $\text{Li}^+$  plating/stripping behavior, which is contributed by the boron moieties immobilizing anions, leading to a dendrite-free Li anode (Scheme 1b). With all these advantages, a solid-state Li metal battery based on the P(V-B) electrolyte has demonstrated excellent long-term cycling stability, superior rate performance, attractive mechanical stability and good flexibility.



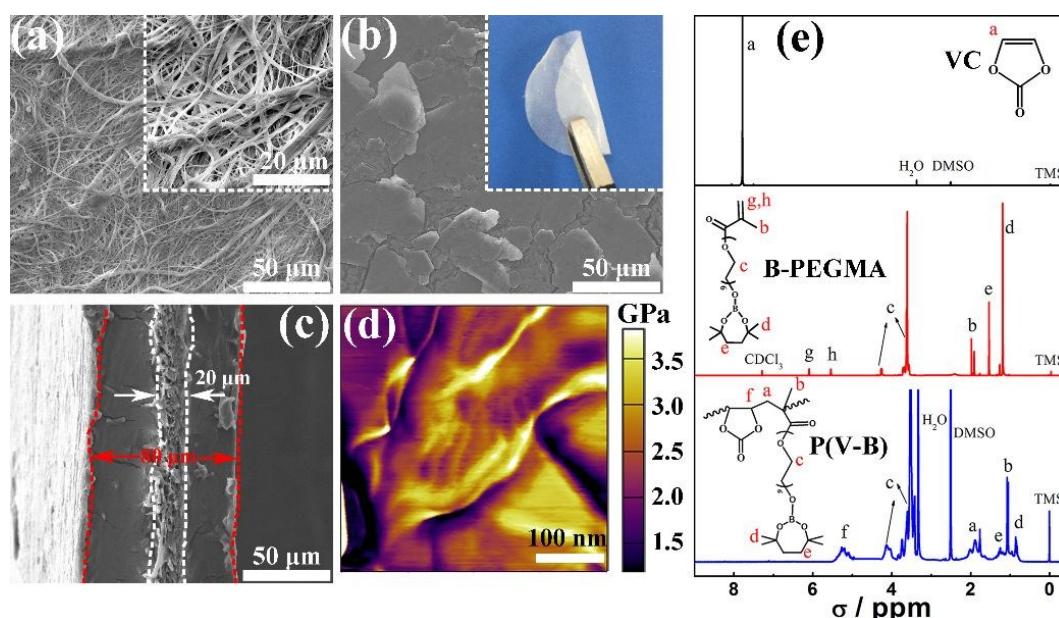
**Scheme 1.** (a) Illustration of the fabrication of P(V-B) electrolyte via in-situ polymerization. (b) Schematic of Li anode structure evolution with LE and P(V-B) electrolyte during cycling.

## Results and discussion

The P(V-B) SPE was synthesized via in-situ polymerization of a solution precursor obtained by mixing LiTFSI, VC and B-PEGMA organic monomers. A cellulose membrane (CM), consisting of copious irregularly shaped holes formed by randomly arranged nanofibers (Fig. 1a), was selected as the mechanical framework. Meanwhile, the high porosity and enriched oxygen functional groups (Fig. S1, ESI†) endow the CM with excellent affinity for organic solvents<sup>26</sup>. After polymerization, the solution precursor is transformed into a solid polymer electrolyte (Fig. S2), and the obtained P(V-B) electrolyte exhibits appreciable flexibility (Fig. 1b). Cross-section scanning electron microscopy (SEM) image (Fig. 1c) shows that the electrolyte components uniformly cover both sides of the CM framework and infiltrate evenly into the internal porous. The <sup>1</sup>H nuclear magnetic resonance (NMR) spectra in Fig. 1e show a new signal (marked as f) that appears at 5.2 ppm in addition to the disappearance of the vinyl groups (marked as a and g, h) from VC and B-PEGMA, demonstrating the formation of the final P(V-B) copolymer, which was further verified by Fourier transform infrared spectroscopy (Fig. S3). The differential scanning calorimetry (DSC) analysis (Fig. S4a) shows no melting peak over the temperature range from -60 to 60 °C, which reveals the amorphous nature of the P(V-B) electrolyte along with the XRD pattern (Fig. S4b). Specifically, the glass



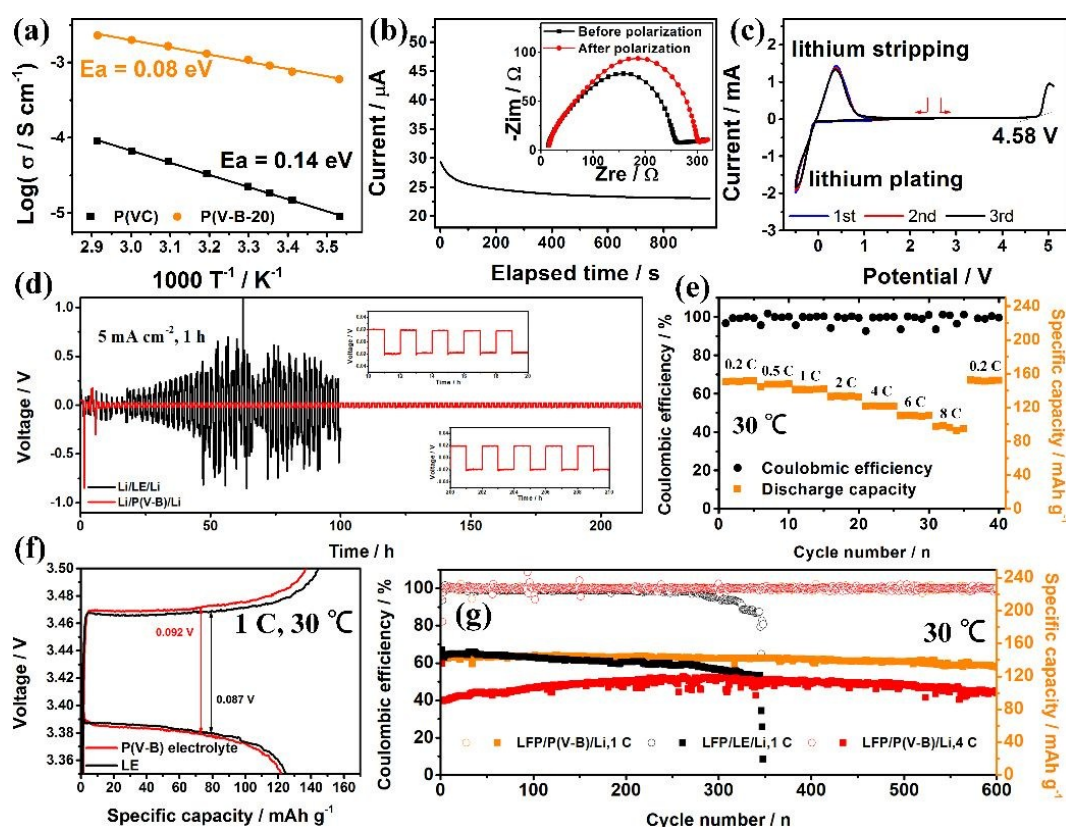
transition temperature ( $T_g$ )  $\sim 22.5$  °C for poly(vinyl chloride) (P(VC)) electrolyte decreases to  $\sim 2.3$  °C in the P(V-B) electrolyte, indicating its superior chain mobility and lower activation energy ( $E_a$ ) for  $\text{Li}^+$  transport<sup>27</sup>. The P(V-B) electrolyte also exhibits desirable mechanical properties with an average Young's modulus of 2.41 GPa (Fig. 1d) and a maximum tensile stress of 9.6 MPa (2.3 MPa for CM, Fig. S5), which can effectively alleviate Li dendrite penetration<sup>28</sup>. Thermal degradation of the P(V-B) electrolyte occurs at  $\sim 277$  °C, confirming its superior thermal and dimensional stability (Fig. S6).



**Fig. 1.** (a) SEM image of the surface morphology of bare CM. (b) Top-view SEM image of P(V-B) electrolyte with optical photograph (inset) and (c) Cross-sectional SEM image of P(V-B) electrolyte. (d) Young's modulus mapping of P(V-B) electrolyte and (e) <sup>1</sup>H NMR spectra of VC, B-PEGMA and as synthesized P(V-B).

The Arrhenius plots in Fig. 2a, S7 reveal that the P(V-B) electrolyte possesses the highest ionic conductivity of  $9.11 \times 10^{-4} \text{ S cm}^{-1}$  with a VC/B-PEGMA ratio of 4/1, which is about 50 times that of the P(VC) electrolyte ( $1.83 \times 10^{-5} \text{ S cm}^{-1}$ ), and is superior to the values reported in the literatures (Table S1, ESI†)<sup>27, 29-35</sup>. Simulation results on the electron cloud density distribution of the molecular model show that the oxygen atoms in both the C=O and the C-O-C groups are negatively charged, which verifies that the improved ionic conductivity should be

ascribed to the enhanced interactions between  $\text{Li}^+$  and the highly mobile EO groups with lower natural bond orbital (NBO) charge (Fig. 3a). Moreover, the conductivity exhibits an Arrhenius-type behavior with temperature, and the decreased value of the activation energy ( $E_a$ ) (Table S2) also demonstrates that the introduced flexible B-PEGMA segment is effective in reducing the energy barrier for  $\text{Li}^+$  transmission<sup>36</sup>.



**Fig. 2.** (a) Arrhenius plots at temperatures from 10 to 70 °C for P(VC) and the P(V-B) electrolyte. (b) Current-time plot of a symmetric Li/P(V-B) electrolyte/Li cell and AC impedance spectra (inset) before and after polarization. (c) Cyclic voltammograms for the first 3 cycles of the P(V-B) electrolyte. (d) Voltage profiles of Li plating/stripping in symmetric Li cells containing LE (black) and P(V-B) electrolyte (red) at current densities of  $5 \text{ mA cm}^{-2}$  for 1 h, with the insets showing enlargements of the indicated time periods. (e) Rate capability for LFP/P(V-B) electrolyte/Li battery. (f, g) Comparison of charge and discharge voltage gap (f) and cycling performance (g) for LFP/Li metal batteries with LE and the P(V-B) electrolyte at 30 °C.

The  $\text{Li}^+$  transference number ( $t_{\text{Li}^+}$ ) of the P(V-B) electrolyte was calculated to be approximately 0.68, which is higher than the values of 0.43 and 0.26 for the P(VC) electrolyte and the LE, respectively (Fig. S8, Table S3). Density functional theory (DFT) calculations,

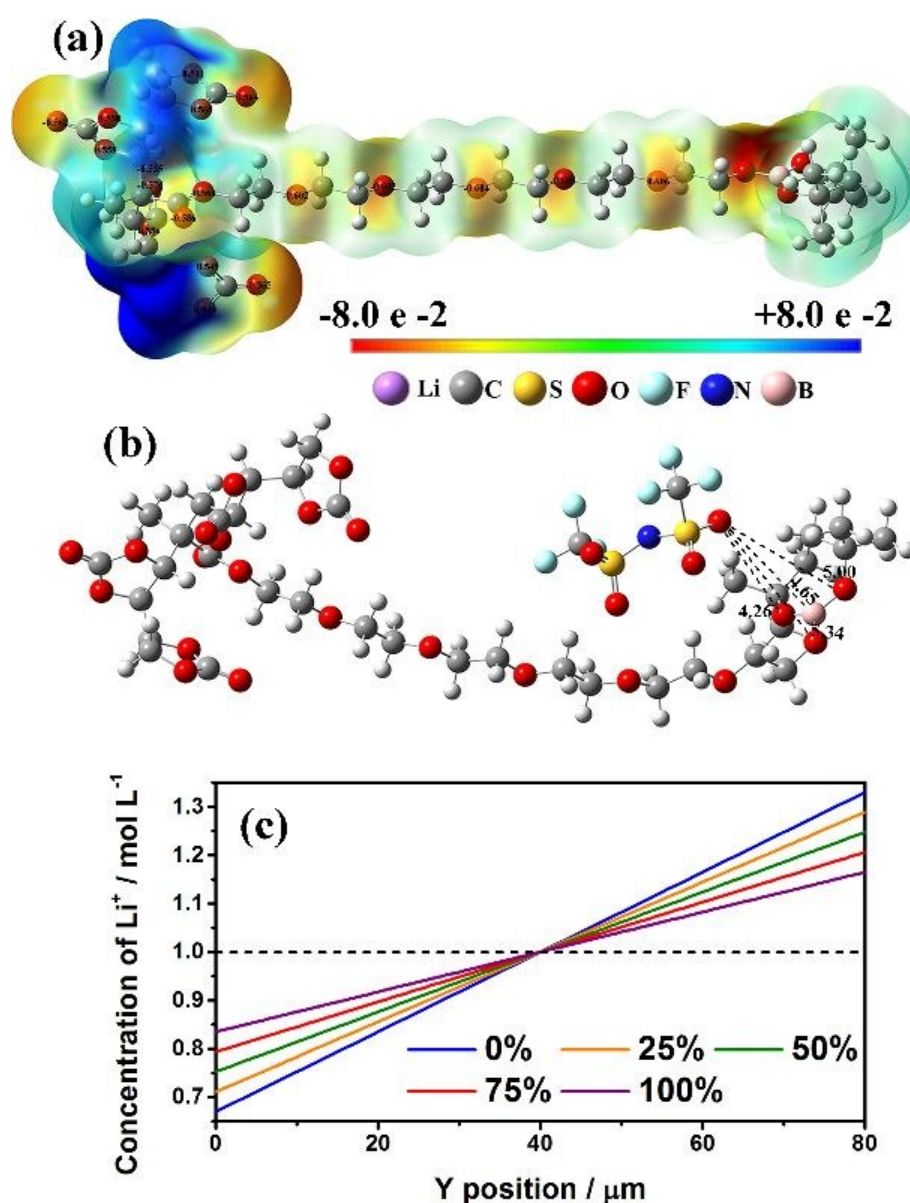


based on the Lee-Yang-Parr correlation energy functional, show that the interaction energy between the boronic ester group and the anion is  $-36.89 \text{ kJ mol}^{-1}$  with a short B-O distance of  $4.65 \text{ \AA}$  (Fig. 3b). All of the above results demonstrate that boron with an empty *p*-orbital can immobilize anions by electronic interaction<sup>37, 38</sup>. More importantly, anion immobilization is beneficial for decreasing the  $\text{Li}^+$  concentration gradient from the bulk electrolyte to the anode surface (Fig. 3c), which could reduce the electric field at the metal electrode and further lead to uniform Li deposition behavior<sup>22, 39</sup>. Moreover, compared to the low oxidative decomposition potential ( $< 4.5 \text{ V}$ ) of conventional  $\text{LiPF}_6$ -based carbonate electrolytes<sup>40</sup>, the P(V-B) electrolyte is electrochemically stable within  $4.58 \text{ V}$  vs.  $\text{Li/Li}^+$  and displays a pair of highly reversible lithium redox peaks (Fig. 2c).

The electrochemical behavior of dynamic  $\text{Li}^+$  plating/stripping was detected by galvanostatic polarization tests. Symmetric Li cells using the P(V-B) electrolyte have a smaller polarization potential and more stable long-term cycling without over-potential increments than that of LE based cells (Fig. 2d, S9a-c). Its excellent performance reveals that the P(V-B) electrolyte could enable outstanding interfacial compatibility with Li metal and facilitate the formation of an electrochemically sustainable SEI film, which is also confirmed by the relatively constant interface impedance over time (Fig. S10).

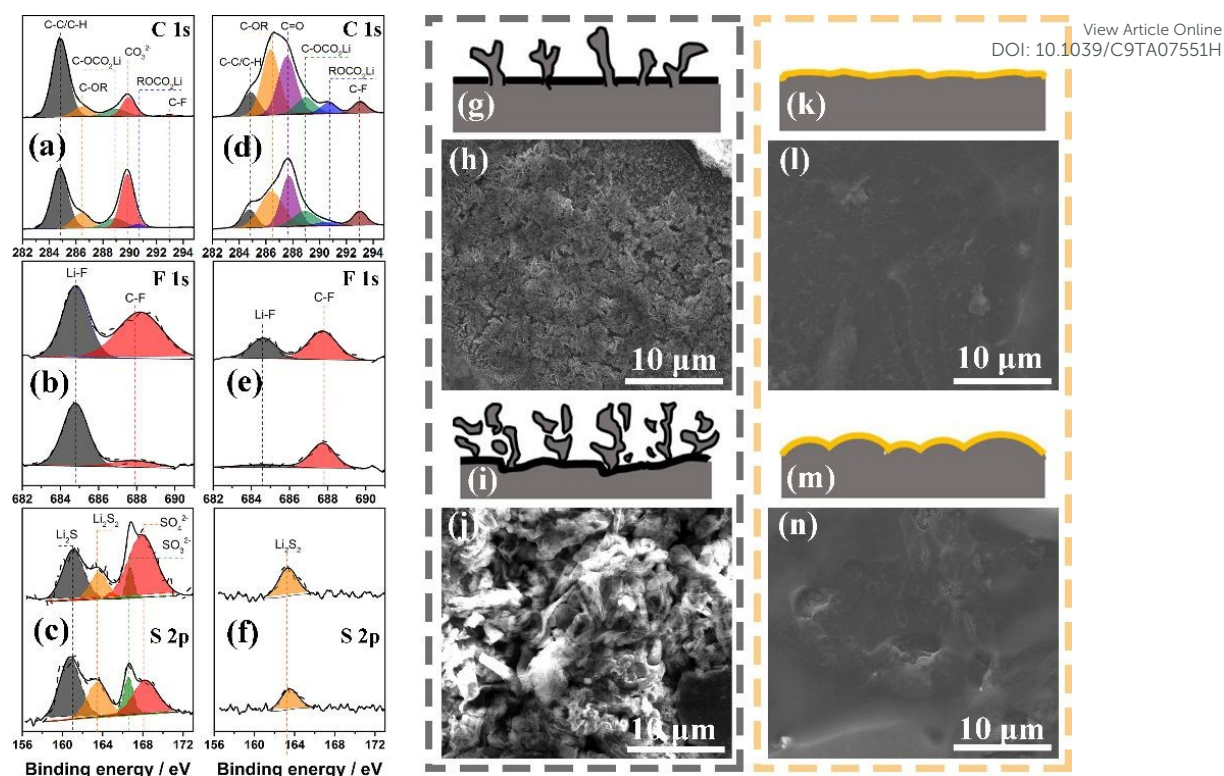
$\text{LiFePO}_4$  (LFP)/Li metal batteries were fabricated to further evaluate the feasibility of the P(V-B) electrolyte. At  $30^\circ\text{C}$ , owing to its high ionic conductivity and low interfacial resistance (Fig. S10a), the discharge capacity of such a battery can remain at  $98.3 \text{ mAh g}^{-1}$  at  $8 \text{ C}$  (Fig. 2e, S11b). Importantly, the polarization between the charge/discharge curves is  $0.092 \text{ V}$  at  $1 \text{ C}$  (Fig. 2f), which is comparable to that with LE, and the capacity retention for the LFP/P(V-B) electrolyte/Li battery is up to  $93.3\%$  after 600 cycles with respect to the initial capacity of  $141.2 \text{ mAh g}^{-1}$ . Even under a high current density of  $4 \text{ C}$ , the P(V-B) based LFP/Li battery still delivers a discharge capacity of  $102.2 \text{ mAh g}^{-1}$  after 600 cycles. For the LE based battery, a short circuit appeared along with reduced Coulombic efficiency upon cycling (Fig. 2g). It is remarkable that

such exceptional battery performance can also be achieved in a large temperature range from 10 °C to 60 °C (Fig. S12), further suggesting the electrochemical stability and attractive interfacial kinetic behavior of the P(V-B) electrolyte. In addition, we also test the P(V-B) electrolyte in a  $\text{LiNi}_{0.6}\text{Co}_{0.2}\text{Mn}_{0.2}\text{O}_2/\text{Li}$  battery with a higher cut-off voltage, and the reversible capacity stabilizes at 115 mAh g<sup>-1</sup> after 100 cycles at 30 °C and 1 C (Fig. S13), highlighting its outstanding interface compatibility with high-voltage cathode materials.



**Fig. 3.** (a-b) Geometrical structure of model unit in P(V-B) with probable electron cloud density distribution (a) and binding energy of anions to the boronic ester groups (b). (c) Simulation of Li<sup>+</sup> distribution from the Li anode surface to the bulk electrolyte at steady state with various proportions of immobilized anions.

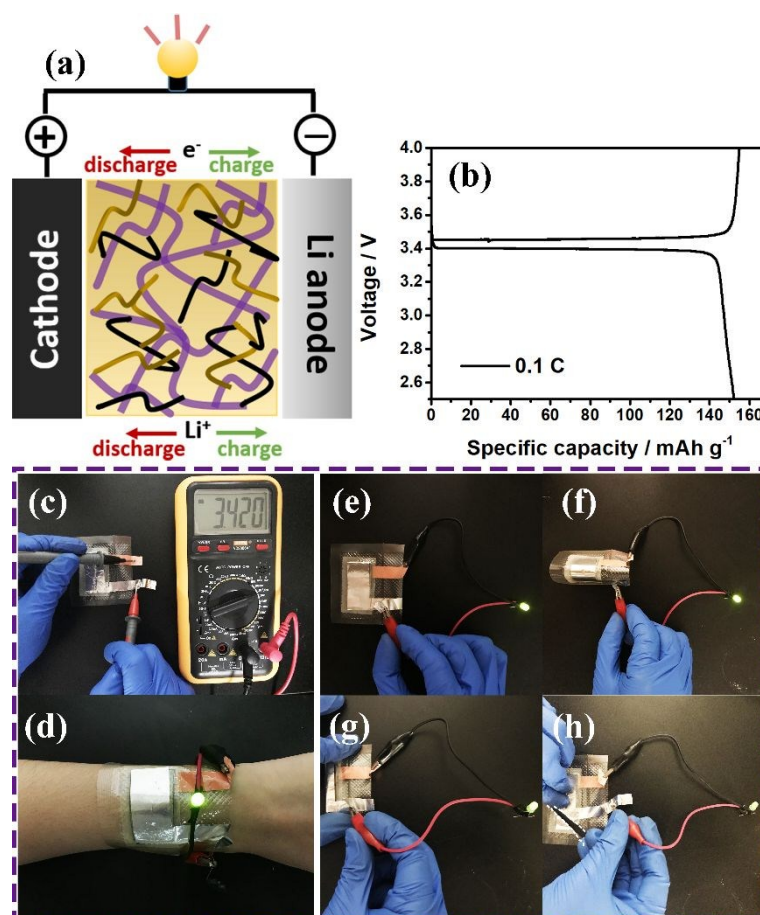
The interfacial properties of the cycled Li anodes taken from LFP/Li metal batteries with LE and the P(V-B) electrolyte was studied. High-resolution X-ray photoelectron spectra (XPS) show that a SEI film comprised of decomposed organics and massive inorganic species ( $\text{Li}_2\text{CO}_3$ , LiF and  $\text{Li}_x\text{S}_y/\text{Li}_x\text{SO}_y$ ) formed on the surface of the Li anode after the initial cycling in the LE based cells, and the chemical constituents change dramatically in the following cycles, accompanying with the constant consumption of Li metal and electrolyte (Fig. 4a-c). In contrast, the organic species with high ion transfer kinetics primarily constitute the pristine native SEI layer for the LFP/Li battery with the P(V-B) electrolyte. In particular, there are no obvious changes in the concentrations of the organic species, while the proportion of the inorganic phases (LiF,  $\text{Li}_x\text{S}_y$ ) decreases on the surface of the Li anode after 50 cycles (Fig. 4d-f). SEM images reveal that, with LE electrolyte, dendritic Li cluster whiskers partially cover the anode surface initially (50 cycles, Fig. 4g, h, S14a, b) and then completely cover the anode surface (100 cycles, Fig. 4i, j, S14 c, d). The whiskers then break from their roots to become isolated Li whiskers (known as dead Li), culminating in a mossy, porous and highly resistant surface layer, which is well consistent with the performance degradation of the LFP/LE/Li battery. On the contrary, the surface of the Li anode in the P(V-B) based battery is smooth and compact without cracks (Fig. 4k,l S14 e, f), and forms no dendrites even after 100 cycles (Fig. 4m, n, S14 g, h). All the above discrepancies demonstrate the synergy between adequate mechanical strength, decreased concentration polarization, and the stable interface enabled by the P(V-B) electrolyte, which can reduce side reactions and realize uniform  $\text{Li}^+$  electrodeposition. Moreover, such outstanding effects are also demonstrated by the electrochemical performance and interface information of Li/Cu asymmetric cells (see Fig. S15 in Supporting Information for details).



**Fig. 4.** (a-f) XPS spectra of C 1s (a, d), F 1s (b, e) and S 2p (c, f) for Li anodes from LFP/Li metal batteries with LE (a-c) and the P(V-B) electrolyte (d-f) after 1 (top) and 50 cycles (bottom) at 1 C. (g-n) SEM SE images and schematic cross sections of Li anodes from LFP/Li metal batteries with LE (g-j) and P(V-B) electrolyte (k-n) after 50 (g, h, k, l) and 100 cycles (i, j, m, n) respectively.

To demonstrate the attractive flexible characteristics and potential applications of the P(V-B) electrolyte in wearable devices, a single layer soft-package cell with a flexible LFP cathode (Fig. S16) and Li foil as the anode was fabricated and exhibits flat plateaus in both charge and discharge profiles (Fig. 5a, b). It is noteworthy that the cell can light up a commercial green light-emitting diode lamp under different bending and folding orientations even after a corner has been cut (Fig. 5 c-h), indicating that the P(V-B) electrolyte with good mechanical integrity and flexibility holds promise for wearable and other flexible devices.





**Fig. 5.** Schematic illustration of a flexible LFP/Li metal pouch cell (a) and the corresponding galvanostatic discharge/charge profile (b) based on voltage testing (c), and photographs of the pouch cell lighting a LED lamp in the flat, and bent states (d-g), and even after having a corner cut off (h).

## Conclusion

In summary, we have demonstrated an in-situ strategy to construct a borate decorated anion-immobilized SPE for all solid state LMBs, which achieves adequate mechanical strength, superior ionic conductivity and enhanced Li<sup>+</sup> transference number. With the high-performance P(V-B) electrolyte, the assembled LMB exhibits long-term cycling stability without dendrite formation on the Li anode at various temperatures (10-60 °C). Simulation and characterization results indicate that the stable performance is attributable to the in-situ formation of an electrochemically stable interface which eliminates inhomogeneous dendrite nucleation, anion-immobilization by borate groups to regulate non-uniform deposits and the adequate mechanical strength to suppress dendritic Li growth. This combination of innovative synthesis design and



facile fabrication provides new insights into interfacial stabilization, Li dendrite inhibition and eventual practical applications of high-energy and safe LMBs.

### Conflicts of interest

There are no conflicts to declare.

### Author contributions

C. M. and W. F. W. conceived the idea and wrote the paper. C. M. and L. Z. and Y. Y. synthesized the materials. C. M., Q. B. X., and L. J. Z., performed the materials tests and characterization. Y. M. F. conducted the density functional theory calculations. F. Z. X. and L. J. Z. provided methods and software for simulation of  $\text{Li}^+$  concentration distribution. M. C., Q. B. X., L. B. C. and W. F. W. performed the data analyses. D. G. I. and D. R. S. provided assistance with interpretation of the data and made revisions to the manuscript. All authors have given approval to the final version of the manuscript.

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